

BRAC UNIVERSITY (SET-A)

CSE110 Exam: Quiz 03

Semester: Spring26

Full Marks: 20

Duration: 20 Mins

Name:	ID:	Section:
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Question 1 [20 Points]

A weekly attendance system stores 7 digits in an array, where **each index** represents a day of the week in order from Saturday to Friday. In this system, classes on **Saturday** (index 0) and **Tuesday** (index 3) are conducted **online**, while all other days are conducted **in-person**. A day is considered balanced if the number of elements before it (to the left) with lower values is equal to the number of elements after it (to the right) with higher values.

Write a Java program to determine and print **all** balanced days **only among the in-person days**, and also print the **total** number of such balanced in-person days.

Given Array	Output
<pre>int[] arr = {40, 10, 20, 100, 15, 30, 5};</pre>	Balanced in-person days are: 2 4 6 Total balanced in-person days: 3
Explanation: Monday (index 2, value 20): left lower count = 1 (10), right higher count = 1 (30), ignoring indices 0 and 3 (online). Since both are equal, the day is balanced. Wednesday (index 4, value 15): left lower count = 1 (10), right higher count = 1 (30), ignoring indices 0 and 3 (online). Since both are equal, the day is balanced. Friday (index 6, value 5): left lower count = 0, right higher count = 0, ignoring indices 0 and 3 (online). Since both are equal, the day is balanced.	

Solution:

public class quiz03{		
public static void main(String [] args){		
int [] arr = {40, 10, 20, 100, 15, 30, 5};	2 arrays initialisation	[2 marks]
int [] arr2 = new int[arr.length];		
int count=0;	3 variables	[1.5 marks]
for(int i=0;i<arr.length;i++){	loop	[0.5 marks]
int left=0,right=0;		
if(i==0 i==3){	online day skip	[2 marks]
continue;		
}		
for(int j=i+1;j<arr.length;j++){	right count	[4 marks]
if(!(j==0 j==3)){		
if(arr[j]>arr[i]){		
right++;		
}		
}		
}		
for(int j=i-1;j>0;j--){	left count	[4 marks]
if(!(j==0 j==3)){		
if(arr[j]<arr[i]){		
left++;		
}		
}		
}		
if(left==right){	count increase, new array update	[3 marks]
arr2[count]=i;		
count++;		
}		
}		
System.out.print("Balanced in-person days are: ");	print	[0.5 mark]
for(int i=0;i<count;i++){	print new array	[2 marks]
System.out.print(arr2[i]+" ");		
}		
System.out.println("\nTotal balanced in-person days: "+count);	print	[0.5 mark]
}		
}		